Derivative Tools

Limit Definition of Derivative

$$f'(x) = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h} \text{ or } \lim_{x \to a} \frac{f(x)-f(a)}{x-a}$$

Power Rule

$$f(x) = a \bullet x^{n}$$

$$f'(x) = a \bullet n \bullet x^{n-1}$$

Sum Rule

$$h(x) = f(x) + g(x)$$

$$h'(x) = f'(x) + g'(x)$$

Product Rule

$$h(x) = f(x) \bullet g(x)$$

$$h'(x) = (f'(x) \bullet g(x)) + (f(x) \bullet g'(x))$$

Quotient Rule

$$h(x) = \frac{f(x)}{g(x)}$$

$$h'(x) = \frac{(f'(x) \bullet g(x)) - (f(x) \bullet g'(x))}{(g(x))^{2}}$$

Chain Rule

$$h(x) = f(g(x))$$

$$h'(x) = f'(g(x)) \bullet g'(x)$$

Exponential

$$f(x) = a^{g(x)}$$

$$f'(x) = a^{g(x)} \bullet ln(a) \bullet g'(x)$$

Logarithm

$$f(x) = log_{a}(x)$$

$$f'(x) = \frac{1}{x \bullet ln(a)} \text{ or } \frac{log_{a}(e)}{x}$$

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Derivative Tools

Trigonometric

$f(x) = sin(x), f'(x) = cos(x)$	$f(x) = cos(x), f'(x) = -sin(x)$
$f(x) = tan(x), f'(x) = sec^2(x)$	$f(x) = cot(x), f'(x) = -csc^2(x)$
$f(x) = sec(x),$ $f'(x) = sec(x) \bullet tan(x)$	$f(x) = csc(x),$ $f'(x) = -csc(x) \bullet cot(x)$

Inverse Trigonometric

$f(x) = sin^{-1}(x),$ $f'(x) = \frac{1}{\sqrt{1-x^2}}$	$f(x) = cos^{-1}(x),$ $f'(x) = -\frac{1}{\sqrt{1-x^2}}$
$f(x) = tan^{-1}(x), f'(x) = \frac{1}{1+x^2}$	$f(x) = cot^{-1}(x), f'(x) = -\frac{1}{1+x^2}$
$f(x) = sec^{-1}(x),$ $f'(x) = \frac{1}{\lvert x\rvert \bullet \sqrt{x^2-1}}$	$f(x) = csc^{-1}(x),$ $f'(x) = -\frac{1}{\lvert x)\rvert \bullet \sqrt{x^2-1}}$

Implicit Differentiation (Chain Rule)

Example: If $x + f(x) = 10$, then $1 + f'(x) = 0$

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